A Progressive RPCA Framework for Image Restoration: From Basic Decomposition to Masked Completion

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ABSTRACT

This report studies image restoration under a progressive implementation of Robust Principal Component Analysis (RPCA). Five versions are developed in sequence. The Basic version establishes a grayscale low-rank and sparse decomposition pipeline. The Color version extends the same decomposition logic to red-green-blue (RGB) images by processing three channels separately. The GUI_advanced version does not alter the optimization model, but improves the experimental interface through background computation, progress display, interactive inspection, and result export. The TV_Regularization version introduces total variation (TV) regularization on the low-rank component so that local oscillation can be suppressed while large-scale structure is retained. The Masked version further modifies the fidelity constraint by excluding manually selected damaged regions from direct fitting and completing them from the remaining valid observations.

1. Introduction

1.1. Problem Background

Low-rank decomposition is a standard tool in matrix analysis and data restoration. When an image is arranged as a matrix, the structural part of the image usually contains strong correlation and redundancy, while scratches, stains, impulsive noise, or local occlusion often appear as relatively sparse corruption. This observation leads naturally to Robust Principal Component Analysis (RPCA), which decomposes an observed matrix into a low-rank part and a sparse part. For this assignment, the mathematical model determines how corrupted data are separated and reconstructed, whereas the interface determines how conveniently the model can be tested, visualized, and compared.

1.2. Task Restatement

The assignment requires a complete implementation of low-rank decomposition for image restoration, with RPCA as the mandatory baseline. Starting from this requirement, the implementation is organized into five progressive versions:

1. **Basic**: grayscale RPCA with a minimal interactive interface;
2. **Color**: extension from grayscale matrices to red-green-blue (RGB) images;
3. **GUI_advanced**: graphical user interface (GUI) enhancement for visualization and usability;
4. **TV_Regularization**: total variation (TV) regularization added to the low-rank component;
5. **Masked**: restoration under partially ignored or known damaged regions.

The central purpose of the report is therefore not only to describe one model, but also to explain how each new version is obtained from the previous one through explicit mathematical or system-level modifications.

1.3. Work Summary

The completed framework follows two parallel lines of development.

- On the modeling side, the method evolves from baseline RPCA to color RPCA, then to TV-regularized RPCA, and finally to masked TV-regularized RPCA.

- On the system side, the interface evolves from a minimal image-selection panel to a multi-functional experimental tool with background computation, interactive inspection, progress display, and result export.

The resulting progression can be written as

$$\mathcal{F}_1 \subset \mathcal{F}_2 \subset \mathcal{F}_3 \subset \mathcal{F}_4 \subset \mathcal{F}_5,$$

where $\mathcal{F}_k$ denotes the function set of Version k.

2. Mathematical Model and Overall Design

2.1. Image Representation and Preprocessing

For a grayscale image, the observation is written as a matrix

$$A \in \mathbb{R}^{m\times n}.$$

For a color image, the observation is treated as a three-channel tensor

$$A \in \mathbb{R}^{m\times n\times 3},$$

and the red, green, and blue channels are denoted by $A^{(R)}$, $A^{(G)}$, and $A^{(B)}$, respectively. In all versions, pixel values are normalized into $[0, 1]$ before optimization so that thresholding and visualization remain numerically stable.

In the masked version, an observation mask

$$M \in \{0, 1\}^{m\times n}$$

is introduced, where $M_{ij} = 1$ indicates that pixel (i, j) is retained in the data-consistency constraint and $M_{ij} = 0$ indicates a user-marked damaged location.

2.2. Baseline Decomposition Objective

The starting point is the classical RPCA model. The observed image is decomposed into a low-rank term L and a sparse term S:

$$A = L + S.$$

The optimization objective is

$$\min_{L,S} \ \|L\|_* + \lambda \|S\|_1 \quad \text{s.t.} \quad A = L + S, \tag{1}$$

where $\|L\|_*$ is the nuclear norm and $\|S\|_1$ is the entrywise ℓ_1 norm. In this formulation, L captures the dominant image structure and S collects localized corruption.

2.3. Optimization Framework

The solver in all model-based versions follows an augmented Lagrangian method (ALM). The basic augmented objective can be written as

$$\mathcal{L}_\mu(L, S, Y) = \|L\|_* + \lambda \|S\|_1 + \langle Y, A - L - S\rangle + \frac{\mu}{2} \|A - L - S\|_F^2, \tag{2}$$

where Y is the multiplier matrix and $\mu > 0$ is the penalty parameter. The algorithm alternates between singular-value thresholding for the low-rank term, soft-thresholding for the sparse term, and multiplier updates.

2.4. Working Assumptions and Modeling Scope

- **Assumption 1.** The corruption occupies a relatively small proportion of the image domain. This is the standard sparse-corruption premise required by RPCA.

- **Assumption 2.** In the masked version, the user-selected boxes identify regions that should be removed from the fidelity constraint. This assumption is consistent with the intended use of manual masking.

Other items, such as normalization, RGB channel processing, panel export, or progress monitoring, are implementation settings rather than mathematical assumptions and are treated as design choices in later sections.

2.5. Main Notation

The principal symbols used in the report are summarized in Table 1.

Table 1
Main notation used in the report.

Symbol	Meaning	Symbol	Meaning
A	observed image matrix	L	low-rank component
S	sparse component	M	observation mask
Ω	observed pixel set	λ	sparse regularization weight
β	TV regularization weight	μ	penalty parameter in ALM
Y	Lagrange multiplier	$\|\cdot\|_*$	nuclear norm
$\|\cdot\|_1$	entrywise ℓ_1 norm	$\mathrm{TV}(\cdot)$	total variation seminorm
$\mathcal{S}_\tau(\cdot)$	soft-threshold operator	$\mathcal{D}_\tau(\cdot)$	singular-value threshold operator

3. Stage I: Basic Version as the Baseline Model

3.1. Model Formulation

The **Basic** version works on a grayscale image matrix and uses the baseline RPCA model. Its role is to establish the minimal decomposition pipeline required by the assignment. Once L and S are obtained, the displayed restoration is based primarily on the low-rank part L, while the sparse part S is shown separately for corruption inspection.

3.2. Iterative Updates

Under the augmented Lagrangian framework, the basic version alternates according to

$$L^{k+1} = \mathcal{D}_{1/\mu_k}\left(A - S^k + \mu_k^{-1} Y^k\right), \tag{3}$$

$$S^{k+1} = \mathcal{S}_{\lambda/\mu_k}\left(A - L^{k+1} + \mu_k^{-1} Y^k\right), \tag{4}$$

$$Y^{k+1} = Y^k + \mu_k (A - L^{k+1} - S^{k+1}). \tag{5}$$

The stopping criterion is based on the normalized residual

$$\varepsilon_{k+1} = \frac{\left\|A - L^{k+1} - S^{k+1}\right\|_F}{\|A\|_F}. \tag{6}$$

3.3. System Characteristics

The baseline interface contains only the functions needed to run a complete RPCA workflow: image selection, parameter input, decomposition execution, and four-panel display. This version establishes the minimum working connection between the mathematical model and the experimental interface.

3.4. Recorded Output for the Basic Version

The currently recorded console output of the Basic version is summarized in Table 2.

Table 2
Recorded numerical output of the Basic version.

Quantity	Value
Image	`1.png`
λ	0.010000
Iterations	35
Final error	6.603478×10^{-8}
$\mathrm{rank}(L)$	20
Sparse ratio of S	0.7831

4. Stage II: Color Extension Compared with the Basic Version

4.1. Main Modification

The **Color** version extends the data domain from one matrix to three aligned color channels. The objective is not changed structurally; instead, the single-channel solver is applied independently to each RGB channel.

4.2. Channel-Wise Model

For channel $c \in \{R, G, B\}$, the decomposition is written as

$$A^{(c)} = L^{(c)} + S^{(c)}.$$

Each channel solves

$$\min_{L^{(c)}, S^{(c)}} \left\| L^{(c)} \right\|_* + \lambda \left\| S^{(c)} \right\|_1 \quad \text{s.t.} \quad A^{(c)} = L^{(c)} + S^{(c)}. \tag{7}$$

After solving the three channel problems, the outputs are stacked as

$$L = \text{stack}\left(L^{(R)}, L^{(G)}, L^{(B)}\right), \qquad S = \text{stack}\left(S^{(R)}, S^{(G)}, S^{(B)}\right).$$

4.3. Increment Relative to the Previous Version

The main differences between the Basic and Color versions are summarized in Table 3.

Table 3
Main differences between the Basic and Color versions.

Aspect	Basic	Color
Data type	grayscale matrix only	grayscale or RGB image
Optimization unit	one matrix	three matrices processed independently
Reported rank	scalar rank	channel-wise rank list
Reported sparsity	scalar sparse ratio	average sparse ratio over channels
Visual result	grayscale restoration	color restoration with channel-consistent display

4.4. Recorded Outputs and Numerical Observation

The outputs of the Basic and Color versions on the same test image are listed together in Table 4.

Table 4
Recorded outputs of the Basic and Color versions.

Quantity	Basic	Color
λ	0.010000	0.010000
Iterations	35	35
Final error	6.603478×10^{-8}	7.462930×10^{-8}
$\text{rank}(L)$	20	$[20, 22, 25]$
Sparse ratio of S	0.7831	0.8361

From the recorded output, three observations can already be made. First, extending the decomposition from one channel to three channels does not noticeably damage numerical stability, since both versions converge in 35 iterations and both final residuals remain at the order of 10^{-8}. Second, the three channels exhibit different effective ranks, which indicates that the structural complexity of the RGB image is not uniform across channels. Third, the sparse ratio increases from 0.7831 to 0.8361, suggesting that after channel-wise decomposition more local detail and corruption are allocated to the sparse component. At the current stage, this supports the conclusion that the Color version increases representational flexibility while preserving stable convergence.

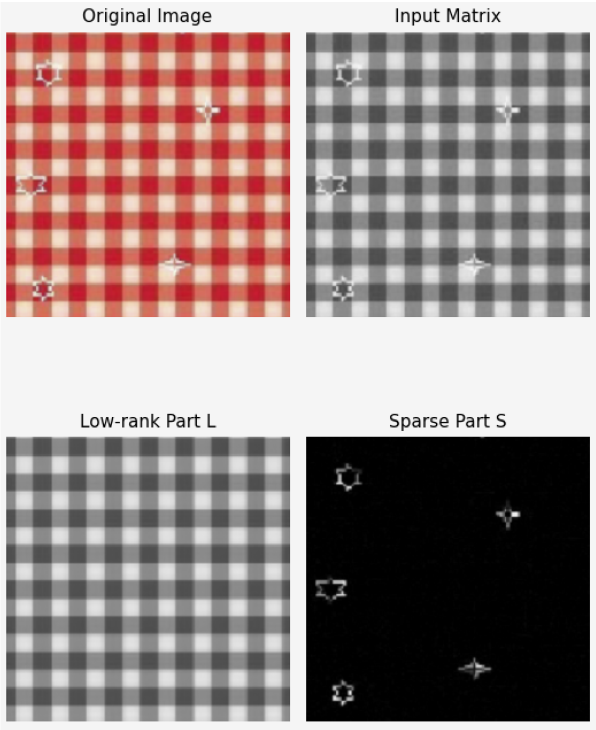

Figure 1: Visual result of the Basic version.

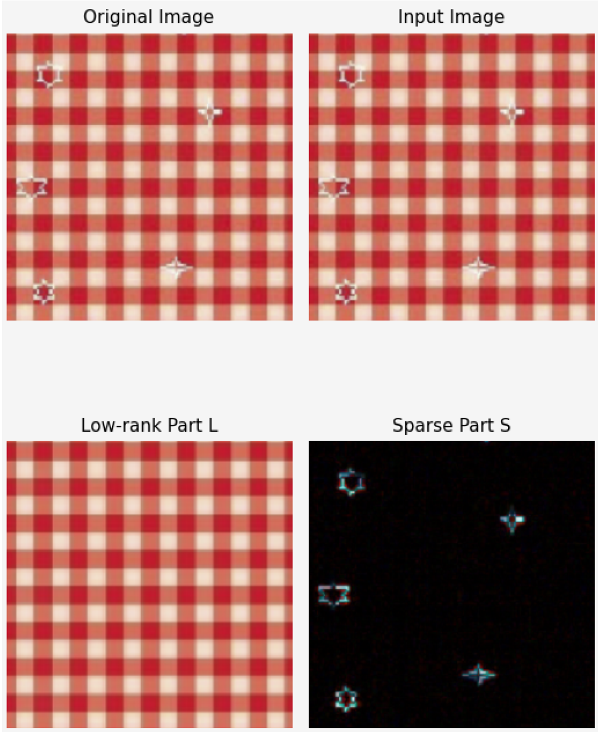

Figure 2: Visual result of the Color version.

5. Stage III: GUI_advanced Compared with the Color Version

5.1. Role of This Stage

The **GUI_advanced** version does not modify the optimization objective. Its function is to turn the Color version from a runnable program into a practical experimental interface. Therefore, this stage is mainly a system-design increment rather than a model-design increment.

5.2. System Improvements

The main changes are listed in Table 5.

Table 5
Main system-level improvements introduced in GUI_advanced.

Aspect	Color version	GUI_advanced
Computation mode	foreground execution	background thread, reduced interface blocking
View control	fixed display	zoom, pan, fit-view, scale control
Result export	no detailed export support	save figure and save individual panels
Runtime feedback	final text output only	per-channel progress, status animation, metadata display
Image source	local folder scan	folder scan plus explicit file browsing

5.3. Design Rationale

Because singular value decomposition (SVD) is the dominant computational cost in RPCA, the interface may become unresponsive during decomposition if the solver runs directly in the main thread. The advanced version therefore separates the computation thread from the interface thread and uses a queue-based progress update mechanism. This design does not change the asymptotic optimization cost, but it significantly improves usability.

5.4. Recorded Output and Numerical Observation

The numerical output of GUI_advanced is shown in Table 6.

Table 6
Recorded output of GUI_advanced.

Quantity	GUI_advanced
Loaded path	`.../GUI_advanced/figs/1.png`
λ	0.010000
Iterations	35
Final error	7.462930×10^{-8}
rank(L)	$[20, 22, 25]$
Sparse ratio of S	0.8361

Comparing this output with the Color version shows that the numerical results are identical: the same λ, the same iteration count, the same final error, the same rank vector, and the same sparse ratio are obtained. Therefore, the contribution of GUI_advanced is not a change in the mathematical solution, but a change in the experimental workflow. This confirms that the advanced GUI improves interaction, runtime monitoring, and export convenience without altering the decomposition itself.

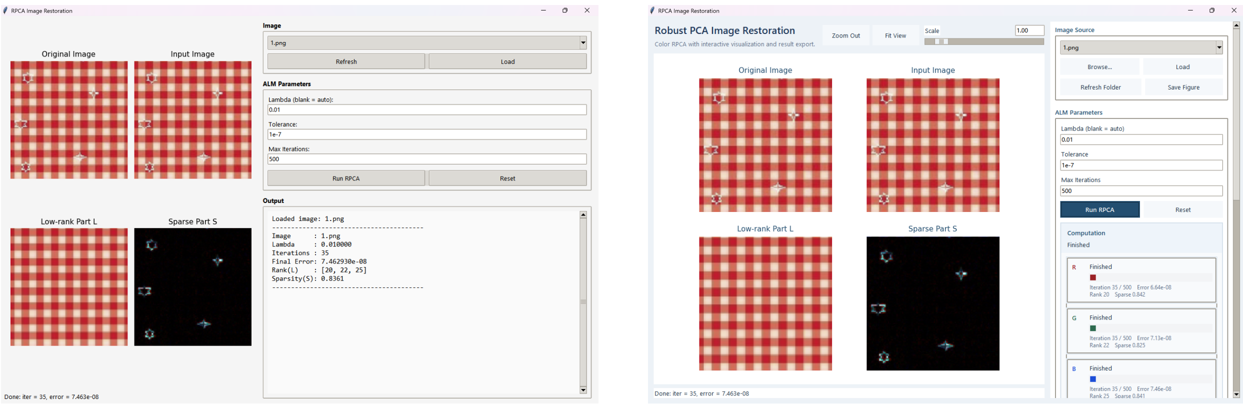

Figure 3: Visual comparison between the Color GUI and GUI_advanced.

6. Stage IV: TV_Regularization Compared with GUI_advanced

6.1. Why TV Regularization Is Added

The previous versions rely on low rank and sparsity alone. In practice, these priors are often insufficient to suppress local oscillation or small isolated artifacts in the recovered low-rank image. The **TV_Regularization** version therefore introduces spatial regularity into the low-rank component.

6.2. Extended Objective

The optimization model becomes

$$\min_{L,S} \; \|L\|_* + \lambda \|S\|_1 + \beta \, \mathrm{TV}(L) \quad \text{s.t.} \quad A = L + S, \tag{8}$$

where $\beta \geq 0$ is the TV weight. For a matrix X, anisotropic first-order finite differences are defined by

$$\nabla_x X(i,j) = X(i,j+1) - X(i,j), \qquad \nabla_y X(i,j) = X(i+1,j) - X(i,j),$$

and the total variation term is written as

$$\mathrm{TV}(X) = \sum_{i,j} \sqrt{\left|\nabla_x X(i,j)\right|^2 + \left|\nabla_y X(i,j)\right|^2}. \tag{9}$$

6.3. Two-Step Update of the Low-Rank Term

The low-rank update is split into two consecutive operations:

$$L^{k+1}_{\frac{1}{2}} = \mathcal{D}_{1/\mu_k}\left(A - S^k + \mu_k^{-1} Y^k\right), \tag{10}$$

$$L^{k+1} = \mathrm{prox}_{(\beta/\mu_k)\mathrm{TV}}\left(L^{k+1}_{\frac{1}{2}}\right). \tag{11}$$

The second step is computed by a primal–dual iteration, which serves as the TV proximal map.

6.4. Increment Relative to the Previous Version

The differences introduced in this stage are summarized in Table 7.

Table 7
Main changes introduced by TV_Regularization.

Aspect	GUI_advanced	TV_Regularization
Objective function	RPCA objective only	RPCA objective plus TV term on L
Low-rank update	singular-value thresholding only	singular-value thresholding followed by TV proximal step
New parameter	none	TV weight β
Expected effect	structural/sparse separation	structural/sparse separation with stronger local smoothness
GUI change	progress and export support	all previous features plus explicit TV parameter control

6.5. Representative Algorithm

The main iteration of the TV-regularized version is summarized in Algorithm 1.

Algorithm 1 Single-channel TV-RPCA by ALM and TV proximal update

Require: Observed matrix A, parameters λ, β, tolerance ε, outer iteration limit K, TV inner iteration limit K_{tv}
Ensure: Low-rank part L and sparse part S

```
Initialize L^0, S^0, Y^0, penalty μ_0
for k = 0, 1, ..., K − 1 do
    Compute L^{k+1}_{1/2} by singular-value thresholding
    Compute L^{k+1} by a TV proximal step with K_tv primal–dual iterations
    Update S^{k+1} by soft-thresholding
    Update Y^{k+1} = Y^k + μ_k(A − L^{k+1} − S^{k+1})
    Compute residual ε_{k+1}
    if ε_{k+1} < ε then
        break
    end if
end for
return L, S
```

6.6. Recorded Outputs and Numerical Observation

Two recorded outputs with different TV weights are listed in Table 8.

Table 8
Recorded outputs of TV_Regularization.

Quantity	$\beta = 0.03$	$\beta = 0.50$
λ	0.010000	0.010000
Iterations	35	34
Final error	6.939658×10^{-8}	9.728158×10^{-8}
rank(L)	$[19, 21, 23]$	$[9, 7, 9]$
Sparse ratio of S	0.8475	0.9492

These outputs show a clear dependence on the TV weight. When $\beta = 0.03$, the method preserves stable convergence, and the rank vector is only slightly lower than that of the Color version, while the sparse ratio increases moderately. When $\beta = 0.50$, the rank decreases sharply and the sparse ratio rises to 0.9492, which indicates that a large amount of image detail is forced away from the low-rank term and absorbed by the sparse term. Therefore, the current outputs support the conclusion that moderate TV regularization is useful, whereas an excessively large TV weight leads to obvious over-smoothing.

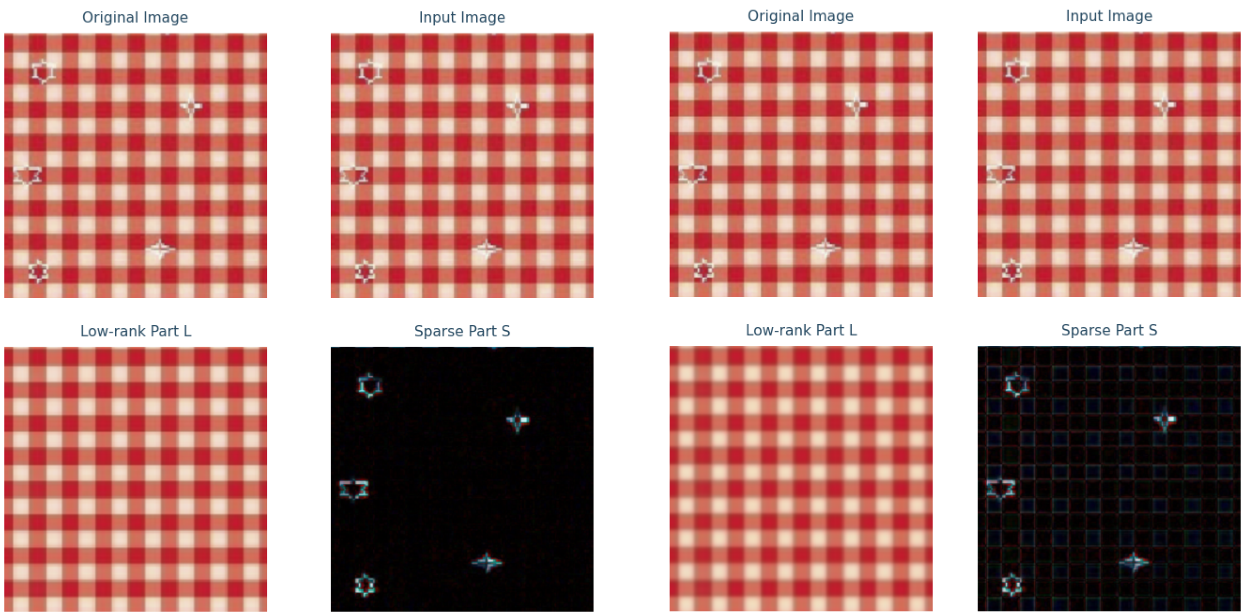

Figure 4: Visual comparison of TV_Regularization under $\beta = 0.03$ and $\beta = 0.50$.

7. Stage V: Masked Version Compared with TV_Regularization

7.1. Motivation

In many damaged-image scenarios, the corrupted region is already known approximately. For such pixels, directly enforcing equality with the observed value is unreasonable because the observation itself is invalid. The **Masked** version therefore modifies the fidelity constraint rather than merely adding another regularizer.

7.2. Masked Observation Model

Let

$$\Omega = \{(i, j) : M_{ij} = 1\}$$

be the observed set. The constrained model becomes

$$\min_{L,S} \|L\|_* + \lambda \|S\|_1 + \beta\, \mathrm{TV}(L) \quad \text{s.t.} \quad \mathcal{P}_\Omega(A) = \mathcal{P}_\Omega(L+S), \tag{12}$$

where $\mathcal{P}_\Omega$ denotes projection onto the observed set. In elementwise form,

$$\mathcal{P}_\Omega(X) = M \odot X.$$

Thus only observed pixels participate in the fidelity term, while masked pixels are inferred from the low-rank and TV priors.

7.3. Masked Updates

Compared with the unmasked TV-RPCA iteration, the main change is that the residual and the temporary update matrices are multiplied by the observation mask. The low-rank half step becomes

$$\widetilde{L}^k = M \odot \left(A - S^k + \mu_k^{-1} Y^k\right) + (1 - M) \odot L^k, \tag{13}$$

followed by

$$L^{k+1} = \mathrm{prox}_{(\beta/\mu_k)\mathrm{TV}}\left(\mathcal{D}_{1/\mu_k}(\widetilde{L}^k)\right). \tag{14}$$

The sparse update is

$$S^{k+1} = \mathcal{S}_{\lambda/\mu_k}\Big(M \odot (A - L^{k+1} + \mu_k^{-1} Y^k)\Big), \tag{15}$$

and the multiplier update uses the masked residual

$$Y^{k+1} = Y^k + \mu_k\, M \odot (A - L^{k+1} - S^{k+1}). \tag{16}$$

The stopping criterion is also evaluated on the observed set only:

$$\varepsilon_{k+1}^{\Omega} = \frac{\left\| M \odot (A - L^{k+1} - S^{k+1}) \right\|_F}{\|M \odot A\|_F}. \tag{17}$$

7.4. Increment Relative to the Previous Version

The differences between TV_Regularization and Masked are summarized in Table 9.

Table 9
Main changes introduced by the Masked version.

Aspect	TV_Regularization	Masked
Fidelity constraint	all pixels satisfy $A = L + S$	only observed pixels satisfy the constraint
New data object	none	observation mask M
Error evaluation	full-image residual	masked residual on Ω
GUI interaction	parameter-based experimentation	parameter control plus manual box selection, undo, and clear
Displayed input	original input image	original image together with observed-region display
Expected effect	denoising and smoothing	denoising, smoothing, and missing-region completion

7.5. System Support for Manual Masking

This stage introduces a direct connection between user interaction and mathematical modeling. A rectangle drawn on the original panel is converted into a binary mask region, and all selected boxes are merged into the global mask matrix M. The observed ratio

$$\eta = \frac{1}{mn} \sum_{i=1}^{m} \sum_{j=1}^{n} M_{ij}$$

is displayed in the interface as an immediate summary of how much valid information remains in the input.

7.6. Recorded Output and Numerical Observation

The manually selected boxes and the final numerical output are listed in Table 10.

Table 10
Recorded mask setting and output of the Masked version.

Quantity	Value
Mask boxes	rows $[56, 152)$, cols $[53, 156)$; rows $[547, 640)$, cols $[368, 489)$
λ	0.010000
TV weight β	0.030000
Observed ratio	0.9599
Iterations	35
Final error	9.771058×10^{-8}
$\text{rank}(L)$	$[17, 19, 21]$
Sparse ratio of S	0.8498

Under an observed ratio of 0.9599, the algorithm still converges within 35 iterations and reaches a final residual of order 10^{-8}. At the same time, the rank vector becomes $[17, 19, 21]$, which is lower than the corresponding ranks in the unmasked Color setting and also below the moderate-TV result $[19, 21, 23]$. This indicates that after known damaged regions are removed from the fidelity constraint, the solution relies more strongly on global low-rank structure and spatial smoothness to infer the missing content. Therefore, the current output supports the conclusion that the Masked version has extended the task from corruption separation to partial-observation completion.

Visual inspection further supports this interpretation. In the original image, five prominent corruption patterns are clearly visible. After two regions are manually masked, the remaining three visible corruption patterns are still successfully identified and separated by the model, while the masked regions are plausibly reconstructed from the surrounding low-rank and smooth structures. This qualitative result is consistent with the numerical evidence above, and further demonstrates that the Masked version is able to preserve corruption detection ability on the observed area while simultaneously performing completion on the missing area.

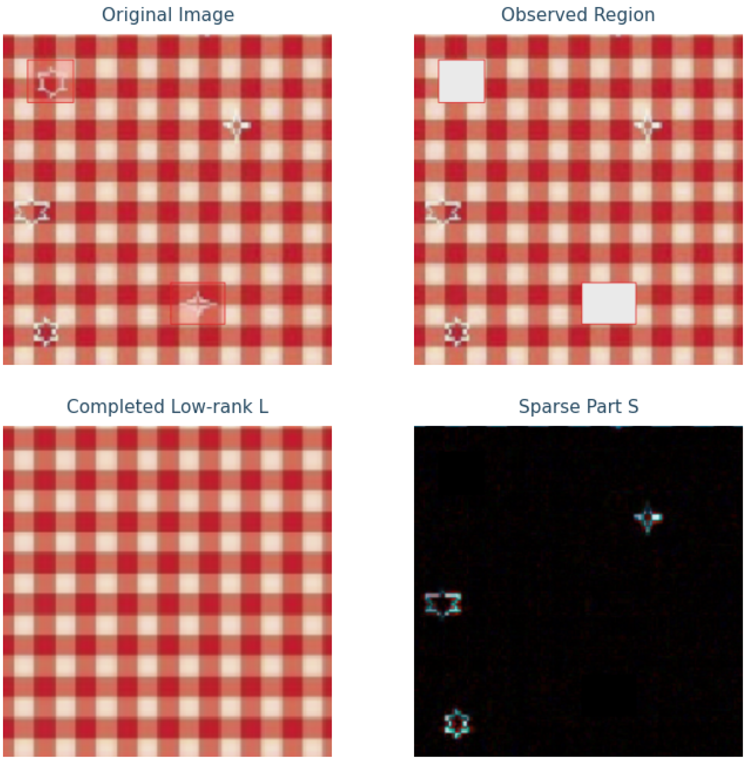

Figure 5: Visual result of the Masked version.

8. Complexity Discussion

Let one image channel have size $m \times n$ and let $p = \min(m, n)$. The dominant cost in the baseline RPCA iteration is the singular-value decomposition, so the per-iteration cost is approximately

$$\mathcal{O}(mnp).$$

For the Color version, the solver is applied to three channels independently, so the total cost is approximately tripled. The GUI_advanced stage does not change optimization complexity, but introduces asynchronous execution and interface-side progress management. The TV_Regularization stage adds a primal–dual inner loop of order

$$\mathcal{O}(K_{\mathrm{tv}} mn)$$

to each outer iteration. The Masked version adds only elementwise mask operations, which contribute about $\mathcal{O}(mn)$ per iteration and do not change the main singular-value-decomposition-dominated order. Therefore the practical complexity ordering is

$$\text{Basic} < \text{Color} \approx \text{GUI\_advanced} < \text{TV\_Regularization} \lesssim \text{Masked}.$$

9. Experimental Record

9.1. Experimental Settings

All results in this section are obtained on the same test image `1.png`. The sparse regularization weight is fixed at $\lambda = 0.01$. For TV_Regularization, two TV weights are tested, namely 0.03 and 0.50. For the Masked version, two rectangular damaged regions are manually selected, and the observed ratio after masking is 0.9599.

Table 11
Experimental settings for the five versions.

Version	Main parameters	Additional setting
Basic	$\lambda = 0.01$	grayscale RPCA
Color	$\lambda = 0.01$	RGB channel-wise RPCA
GUI_advanced	$\lambda = 0.01$	same solver as Color
TV_Regularization	$\lambda = 0.01,\ \beta \in \{0.03, 0.50\}$	TV proximal step
Masked	$\lambda = 0.01,\ \beta = 0.03$	two mask boxes, $\eta = 0.9599$

9.2. Numerical Comparison

Table 12 summarizes the numerical outputs of the five versions.

Table 12
Numerical outputs of the five versions.

Version	λ / β	Iterations	Final residual	rank(L)	Sparse ratio
Basic	0.01 / –	35	6.603478×10^{-8}	20	0.7831
Color	0.01 / –	35	7.462930×10^{-8}	$[20, 22, 25]$	0.8361
GUI_advanced	0.01 / –	35	7.462930×10^{-8}	$[20, 22, 25]$	0.8361
TV_Regularization ($\beta = 0.03$)	0.01 / 0.03	35	6.939658×10^{-8}	$[19, 21, 23]$	0.8475
TV_Regularization ($\beta = 0.50$)	0.01 / 0.50	34	9.728158×10^{-8}	$[9, 7, 9]$	0.9492
Masked	0.01 / 0.03	35	9.771058×10^{-8}	$[17, 19, 21]$	0.8498

9.3. Numerical Observations

The numerical results lead to the following observations.

1. Compared with the Basic version, the Color version produces a higher-rank low-rank component and a larger sparse ratio, while both methods converge within 35 iterations and reach residuals on the order of 10^{-8}. This shows that extending the model from grayscale to RGB channels increases the expressive capacity of the decomposition without causing convergence difficulties.

2. GUI_advanced gives exactly the same numerical output as the Color version. This is consistent with its role as an interface and workflow improvement rather than a modification of the underlying solver.

3. For TV_Regularization, the choice of β has a clear influence on the result. When $\beta = 0.03$, the decomposition remains close to the Color version, while $\beta = 0.50$ causes a sharp decrease in rank and a noticeable increase in sparse ratio. This suggests that a small TV weight is helpful, whereas an overly large TV weight tends to over-smooth the recovered image.

4. The Masked version still converges in 35 iterations with a residual of order 10^{-8} when the observed ratio is 0.9599. This indicates that the method remains stable after part of the image is removed from the fidelity term, and that the missing regions can be inferred from the remaining observed pixels.

10. Conclusion

This report develops a progressive RPCA-based image restoration framework in five stages. The Basic version provides a valid grayscale decomposition baseline. The Color version generalizes the method to RGB data without damaging convergence stability. The GUI_advanced version improves experimental usability while preserving numerical equivalence with the Color solver. The TV_Regularization version introduces spatial smoothness and shows that the TV weight must be controlled carefully. The Masked version further changes the fidelity constraint so that known damaged regions can be excluded from direct fitting and completed from the remaining observations. According to the current outputs, the five versions form a coherent incremental path from baseline decomposition to partial-observation restoration.